

1 Physics Topics

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1.1 Electrodynamics

1.1.1 Maxwell-Heaviside Equations

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad \oiint_S \mathbf{E} \cdot d\mathbf{a} = \iiint_V \rho dV \quad (\text{Gauss' Law})$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad \oint_C \mathbf{E} \cdot d\mathbf{l} = -\frac{\partial}{\partial t} \iint_S \mathbf{B} \cdot d\mathbf{a} \quad (\text{Maxwell-Faraday Equation})$$

$$\nabla \cdot \mathbf{B} = 0 \quad \oiint_S \mathbf{B} \cdot d\mathbf{a} = 0 \quad (\text{Gauss' Law})$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \quad \oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 \iint_S \mathbf{J} \cdot d\mathbf{a} \quad (\text{Ampere's Law})$$

$$+ \epsilon_0 \mu_0 \frac{\partial \mathbf{E}}{\partial t} \quad + \epsilon_0 \mu_0 \iint_S \frac{\partial \mathbf{E}}{\partial t} \cdot d\mathbf{a} \quad (\text{Maxwell's Correction})$$

1.1.2 Fields in Matter

Fields inside media that are not vacuum:

1.1.3 Retarded Potentials

$$\mathbf{E}(\mathbf{x}, t) = \frac{q}{4\pi\epsilon_0} \left\{ \left[\frac{1}{\gamma^2} \frac{\mathbf{R} - R\boldsymbol{\beta}}{(R - \boldsymbol{\beta} \cdot \mathbf{R})^3} \right]_{\text{ret}} + \left[\frac{1}{c^2} \frac{\mathbf{R} \times [(\mathbf{R} - R\boldsymbol{\beta}) \times \mathbf{a}]}{(R - \mathbf{R} \cdot \boldsymbol{\beta})^3} \right]_{\text{ret}} \right\}$$

With:

$$\gamma = (1 - \beta^2)^{-1/2} \quad \boldsymbol{\beta} = \frac{\mathbf{v}}{c}$$

$$\mathbf{R} = \mathbf{x} - \mathbf{x}_0(\tau_0)$$

1.2 Particle Physics

1.2.1 Gamma Matrix Identities

$$\begin{aligned}
\gamma^{0\dagger} &= \gamma^0 \\
\gamma^{j\dagger} &= -\gamma^j \quad (j = 1, 2, 3) \\
\gamma^0 \gamma^{\mu\dagger} \gamma^0 &= \gamma^\mu \\
\gamma^0 \gamma^0 &= \mathbb{I} \\
\gamma^\mu \gamma_\mu &= 4 \\
\gamma^\mu \gamma_\nu \gamma_\mu &= -2\gamma_\nu \\
\gamma^\mu a_\mu &= \gamma_\mu a^\mu = \not{a} \\
\psi^\dagger \gamma^0 &= \bar{\psi}
\end{aligned}
\qquad
\begin{aligned}
\{\gamma_\mu, \gamma_\nu\} &= \gamma_\mu \gamma_\nu + \gamma_\nu \gamma_\mu = 2g_{\mu\nu} \\
\gamma^\mu \gamma_\mu &= 4 \\
[\gamma^\mu, a_\mu] &= 0 \\
\gamma^\mu a_\mu &= \gamma_\mu a^\mu = \not{a} \\
\text{Tr}(\mathbb{I}) &= 4 \\
\text{Tr}(\gamma_\mu \gamma_\nu) &= \text{Tr}(\gamma^\mu \gamma^\nu) = 4 \\
\text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) &= 4(g^{\mu\nu} g^{\rho\sigma} - g^{\mu\rho} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\rho})
\end{aligned}$$

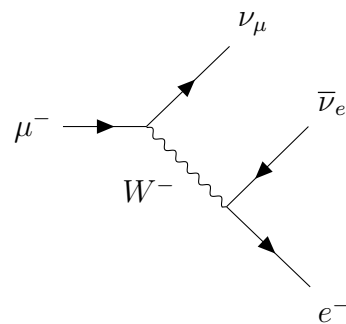
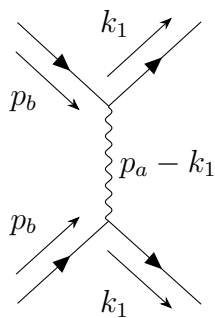
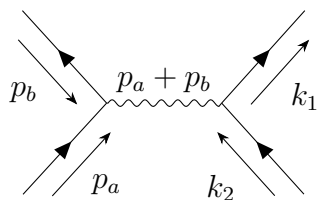
Gamma 5 matrix:

$$\begin{aligned}
\gamma_5^\dagger &= \gamma_5 \\
\gamma_5 \gamma_5 &= \mathbb{I} \\
\{\gamma_5, \gamma_\nu\} &= 0 \\
\text{Tr}(\gamma_5) &= 0 \\
\text{Tr}(\gamma_\mu \gamma_5) &= 0 \\
\text{Tr}(\gamma_\mu \gamma_\nu \gamma_5) &= 0 \\
\text{Tr}(\gamma_\mu \gamma_\nu \gamma_\rho \gamma_5) &= 0 \\
\text{Tr}(\gamma_\mu \gamma_\nu \gamma_\rho \gamma_\sigma \gamma_5) &= 4i\epsilon_{\mu\nu\rho\sigma}
\end{aligned}
\qquad
\begin{aligned}
P_L &= \frac{1 - \gamma_5}{2} \\
P_R &= \frac{1 + \gamma_5}{2} \\
P_L \gamma_\mu &= \gamma_\mu P_R \\
P_L(\gamma^\mu \gamma^\nu \dots \text{odd}) P_L &= 0 \\
P_L(\gamma^\mu \gamma^\nu \dots \text{even}) P_L &= P_L(\gamma^\mu \gamma^\nu \dots)
\end{aligned}$$

Quick Complex Conjugates:

$$\begin{aligned}
(\bar{u} \gamma^\mu v)^* &= \bar{v} \gamma^\mu u \\
(\bar{u} \gamma^\mu P_L v)^* &= \bar{v} \gamma^\mu P_L u
\end{aligned}$$

1.2.2 Mandelstam Variables



$$s = (p_a + p_b)^2 c^2 = (k_1 + k_2)^2 c^2 \quad t = (p_a - k_1)^2 c^2 = (p_b - k_2)^2 c^2$$

1.2.3 Feynman Rules



$\longleftrightarrow$

$u(p, s)$



$\longleftrightarrow$

$\bar{v}(p, s)$



$\longleftrightarrow$

$\bar{u}(p, s)$



$\longleftrightarrow$

$v(p, s)$



$\longleftrightarrow$

$-iQ_f e \gamma^\mu$